

2D Hyperbolic Tensor Calculations

Metric Tensor

$$g_{\mu\nu} = \begin{bmatrix} \frac{\alpha^2}{y^2} & 0 \\ 0 & \frac{\alpha^2}{y^2} \end{bmatrix}$$

Christoffel Symbols

$$\Gamma_{01}^0 = -\frac{1}{y}$$

$$\Gamma_{10}^0 = -\frac{1}{y}$$

$$\Gamma_{00}^1 = \frac{1}{y}$$

$$\Gamma_{11}^1 = -\frac{1}{y}$$

Riemann Tensor

$$R_{101}^0 = -\frac{1}{y^2}$$

$$R_{110}^0 = \frac{1}{y^2}$$

$$R_{001}^1 = \frac{1}{y^2}$$

$$R_{010}^1 = -\frac{1}{y^2}$$

Ricci Tensor

$$R_{00} = -\frac{1}{y^2}$$

$$R_{11} = -\frac{1}{y^2}$$

Ricci Scalar

$$R = -\frac{2}{\alpha^2}$$

Einstein Tensor

$$G_{\mu\nu} = 0$$